STUDENT OPPORTUNITY MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

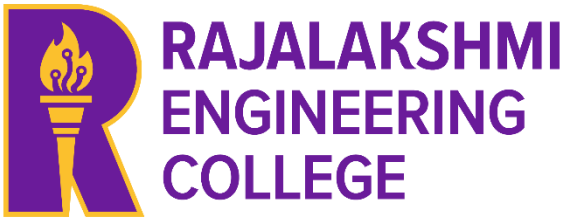
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "Student Opportunity Management System" is the bonafide work of Vidyut Rao S (240701587), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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Submitted for the End Semester Examination (Individual Project) held on _____.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, Vidyut Rao S (240701587), hereby declare that the project report titled “Student Opportunity Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: _____

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Vidyut Rao S

ABSTRACT

The Student Opportunity Management System is a Zoho Creator application for organising student profiles, opportunities, applications, skills, certificates and opportunity matches. Forms capture structured information and lookup controls associate selected records. Reports provide views of saved data. An enabled application-submission workflow contains Deluge logic intended to compare a student's CGPA against an opportunity's minimum. This report presents the design, interface, configuration and proposed tests. The application was inspected in the owner account without submitting records; therefore the workflow's runtime outcomes remain to be tested. The project demonstrates low-code forms, reports, relationships and validation for an academic opportunity-management scenario.

Keywords: Zoho Creator, student profile, opportunity, CGPA, skills, Deluge

PROJECT AT A GLANCE

Attribute	Details
Project	Student Opportunity Management System
Student	Vidyut Rao S — 240701587
Section	CSE-J
Platform	Zoho Creator
Components	Six forms and six report links
Workflow inspected	Validate CGPA Requirement
Event	Applications — validation on form submission
Inspection	Signed-in owner account; read-only
Testing	Navigation observed; submission tests pending

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
CGPA	Cumulative Grade Point Average
CRUD	Create, Read, Update and Delete
FR	Functional Requirement
NFR	Non-functional Requirement
UI	User Interface
URL	Uniform Resource Locator
UAT	User Acceptance Testing

CHAPTER 1
INTRODUCTION

1.1 Background and Problem Statement

Students encounter opportunities with different eligibility requirements. Academic records, skills, certificates and application progress can be spread across separate documents. Repeated copying and manual comparison make it difficult to keep this information consistent and current.

The Student Opportunity Management System organises these items using Zoho Creator. It provides student and opportunity records together with applications, matches, learning goals and certificates. The interface groups each form with its corresponding report.

1.2 Objectives

- Maintain student profiles and opportunity information.
- Record applications and related learning activities.
- Provide fields for match details and certificate records.
- Configure an eligibility check and support report-based review.

1.2.1 Rationale

A structured record makes the relationship between a student and a particular opportunity explicit. A coordinator can review the requirement separately from the application and can correct a source record without copying the same description into every note. This reduces the need to search through messages, but it does not replace judgement about whether an opportunity is appropriate.

The system is intended to organise information for review. An opportunity may also have eligibility conditions that cannot be captured by a single number, such as course restrictions, portfolio requirements or deadlines. Those conditions still need to be read and checked by the user.

1.3 Scope

The selected application has six form and report areas. This report documents their visible fields and one inspected workflow. A field labelled Match Percentage or Extracted Text does not by itself establish automatic calculation or extraction.

1.4 Organisation of the Report

The following chapters cover the system study, requirements, design, implementation, testing, governance, results and conclusion. Appendices provide the configuration, user manual and examiner records.

1.5 Boundaries and Assumptions

The design assumes that a student profile and an opportunity are available before a related application is entered. A user selects these records through lookup controls. Other details, such as the match percentage and missing skills, must be checked for accuracy because the inspection did not determine their source.

The project concerns a Creator application operated through a signed-in account. It does not establish institutional approval, unrestricted public access or an automated decision-making service. The description of each component in later chapters reflects what was visible during the review.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Existing Practice

In a representative manual process, opportunities arrive through notices or email, student information is kept in separate files, and application progress is tracked in a spreadsheet. This is a general problem model, not a measured study of a named organisation.

Table 2.1: Existing and proposed approach

Activity	**Manual**	**Creator application**
Profiles	Separate files	Structured form
Opportunities	Messages and notices	Opportunity form
Applications	Individual tracking	Related application records
Eligibility	Manual comparison	Configured CGPA validation
Learning	Separate notes	Skill Development form

2.1.1 Practical Difficulties

A list maintained in isolation may contain the right opportunity title but omit the latest deadline or a change to the eligibility criteria. A student profile in another file may also use different skill names. These differences make comparisons slow and can lead to a decision based on stale or inconsistent information.

A shared structured system offers a common place to record these details. Its value depends on responsible maintenance: the opportunity owner should check source information, and the student should keep academic and skill details current.

2.2 Platform Concepts

Concept	Purpose	Verification boundary
Forms	Capture structured records	Field presence alone does not prove validation
Lookups	Select a related record	The related record must exist
Reports	Review saved records	Visibility depends on permissions
Workflows	Execute actions on events	Configuration needs runtime testing
Deluge	Express custom logic	Behaviour depends on inputs and execution

Low-code tools combine visual components with programmable actions. They reduce the amount of interface and hosting code needed for a course project, while still requiring careful design and testing.

2.2.1 Applying the Concepts

The forms in this project separate relatively stable information, such as student details, from transaction records, such as an application to a specific opportunity. Lookup fields link the two. Reports then provide a list view for review, while a form event can invoke additional logic.

Low-code configuration should still be treated as software. Field names, event timing, data types and permissions affect the result. A visible workflow is evidence of configuration; a reproducible test is needed to demonstrate what happens with real inputs.

2.3 Proposed System

An authorised user maintains student and opportunity records. Applications and matches can reference both; skill-development and certificate records provide additional context. List reports provide a way to review saved records.

The configured CGPA validation reads values from the selected student and opportunity. Its behaviour should be tested with values below, equal to and above the minimum, and with incomplete data.

2.4 Study Boundary

The inspection does not establish an automatic matching algorithm, automated certificate extraction, notification delivery or production readiness.

2.5 Expected Benefits and Limits

A single navigation structure makes related records easier to locate and provides a consistent view of the application. Structured fields can also make it clearer which information needs review. These are design expectations; no comparative timing or user study was conducted.

The system should be treated as a course project until access, data quality, validation and recovery have been assessed. In particular, a numeric match field should not be presented as an automatically computed recommendation unless the calculation and its results have been verified.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and Functional Requirements

Students need a useful record of their skills and applications. An owner or coordinator maintains opportunities and reviews data. The examiner reviews the work and evidence.

Table 3.1: Functional requirements

ID	Requirement
FR01	Capture student profile and academic details.
FR02	Record opportunity requirements and deadlines.
FR03	Associate applications with students and opportunities.
FR04	Record skill goals and certificate details.
FR05	Record match percentage and related skills.
FR06	Provide a report for each form.
FR07	Configure a CGPA check on application submission.

3.1.1 Requirements Traceability

FR01 and FR02 establish the source records. FR03 and FR05 depend on selecting the appropriate student and opportunity, while FR04 captures related evidence and learning work. FR06 concerns reviewing the stored information, and FR07 concerns a configured eligibility check.

These requirements describe intended functionality. The implementation chapters identify which controls and workflow settings were observed; Chapter 6 separates those observations from scenarios that remain to be executed.

3.2 Non-Functional Requirements

Table 3.2: Non-functional requirements

Area	Requirement
Usability	Provide clear labels and consistent navigation
Privacy	Restrict contact and academic data to authorised users
Quality	Review CGPA scales, deadlines and skills
Maintainability	Use understandable names and review changes
Availability	Account and hosted-service availability affect access
Auditability	Retain appropriate test and change evidence

3.3 Hardware and Software

A compatible computer or device, supported browser, internet connection and authorised Zoho Creator account are needed to access the hosted application. Deluge is used for the inspected validation action.

3.3.1 Operating Conditions

Access to the hosted application depends on a working network connection and a valid account. If either is unavailable, the user may not be able to enter or review records. Availability, response time and recovery were not measured during this project review.

Usability also depends on the quality of the data. Clear labels help users understand fields, but the application owner must still decide who may see contact details and how obsolete opportunities should be handled.

3.4 Use Case — Maintain a Profile

Item	Description
Actor	Authorised application user
Precondition	The user can access the Student Profile form.
Main flow	Enter identity, academic and skill information; submit and review the report.
Alternative	Correct inaccurate or missing details before submission.
Postcondition	A profile is available if submission succeeds.
Evidence	Form and existing report rows observed; new entry not tested.

3.4.1 Profile Review

The student's department, year of study and career goal provide context for interpreting the profile. Technical skills and project summaries are free-text descriptions, so consistent terminology is helpful when a user reviews them against an opportunity. The interface does not establish that every entry has been independently verified.

After a successful submission, the user should review the profile in the corresponding report. If the displayed information is wrong, the correction process should follow the access rules of the deployed application. The present review did not create or edit a profile.

3.5 Use Case — Record an Application

Item	Description
Actor	Authorised application user
Precondition	Student and opportunity records exist.
Main flow	Select student and opportunity; enter date, status and remarks; submit.
Validation	Configured logic compares the selected CGPA and minimum.
Alternative	Review any eligibility alert and correct the source data if needed.
Postcondition	A record is saved only if submission succeeds.
Status	End-to-end outcomes remain to be tested.

3.5.1 Eligibility and Follow-up

A student's recorded CGPA is only one part of the opportunity requirements. The user should also read the description, deadline and eligibility criteria before treating an application as suitable. A validation message should be interpreted in the context of the selected records and their grading scale.

Application Status and Remarks allow progress to be recorded, but their presence does not prove an automatic status transition. A controlled test should confirm the submission result and the contents of the resulting report row.

CHAPTER 4
SYSTEM DESIGN

4.1 Architecture

The Creator web interface is the presentation layer. Forms capture data; lookup controls select related records; and list reports present saved records. Creator executes configured workflow actions at a form event. The application and its records are managed through the Creator account.

Layer	Components
Interface	Forms, navigation and reports
Data	Student, opportunity, application, match, skill and certificate records
Logic	Applications validation using Deluge
Platform	Hosted Zoho Creator application

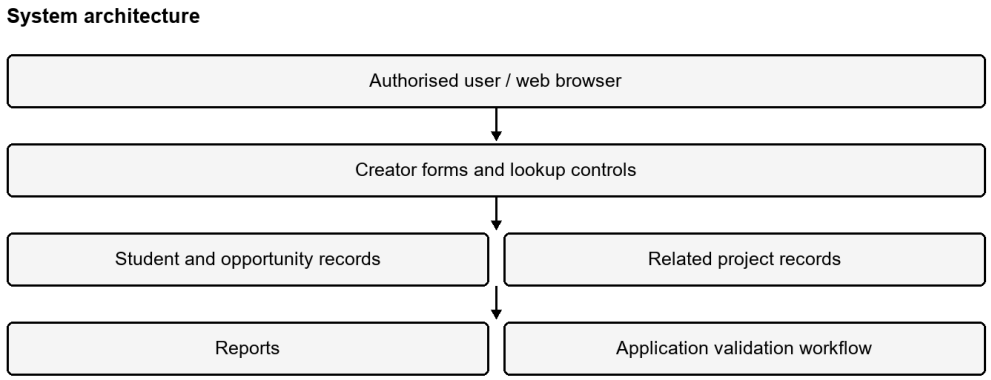


Figure 4.1: Architecture of the Creator application

4.1.1 Layer Responsibilities

The form layer gathers values and lookup selections. Creator stores the records, while list reports provide views of that data. The inspected application workflow runs at validation time and compares two related values; no additional service is shown in this architecture.

This diagram describes the visible component relationships. It is not a network deployment diagram, and it does not assert that a separate server or external matching service has been implemented.

4.2 Data Model

Table 4.1: Form relationships

Form	Related selection	Purpose
Student Profile	—	Student and academic details
Opportunities	—	Opportunity requirements
Applications	Student; Opportunity	Application status
Opportunity Matches	Student; Opportunity	Match and skills
Skill Development	Student; Related Opportunity	Learning goals
Student Certificates	Student	Certificate details

This is a conceptual model based on visible controls. The inspection did not export all database constraints or determine deletion rules.

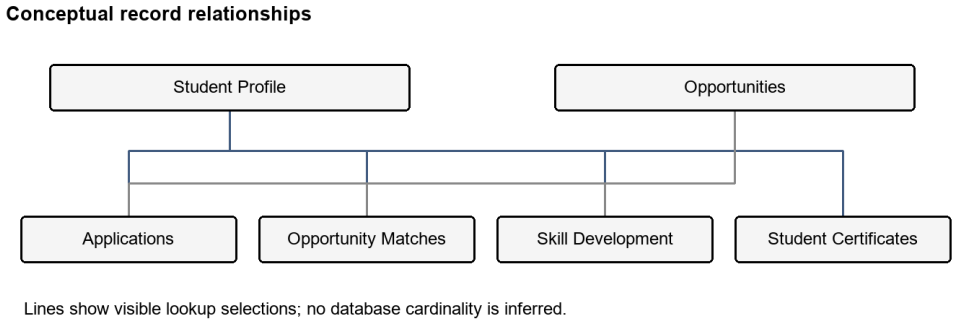


Figure 4.2: Conceptual relationships among project records

4.2.1 Relationship Interpretation

A student can be selected in applications, matches, skill-development entries and certificate entries. An opportunity can be selected in applications, matches and skill-development records. The diagram presents those visible selections conceptually; it does not establish database cardinalities or cascade behaviour.

4.3 Student Profile Design

Table 4.2: Profile fields

Field	Purpose
Full Name; College Email; Phone Number	Identify and contact the student
Department; Year of Study	Academic context
CGPA	Academic value used by the validation
Technical Skills	Describe skills
Certifications Summary; Projects Summary	Summarise experience
Career Goal; Profile Status	Record interest and status

Academic and contact information should be reviewed for accuracy and disclosed only to authorised users.

4.3.1 Profile Field Use

Name and contact fields help identify the person associated with a record. Department and year of study provide educational context. CGPA is particularly important because the observed application-validation action reads that value from the selected student profile.

Skill and project summaries are descriptive text. They should be kept current and written consistently so a reviewer can understand them. Career Goal and Profile Status provide additional context; their presence does not by itself assign access permissions or verify the profile.

4.4 Opportunity Design

Table 4.3: Opportunity fields

Field	Purpose
Opportunity Title; Organization Name	Identify the opportunity
Opportunity Type; Description	Classify and describe it
Required Skills	Record expected skills
Minimum CGPA; Eligibility Criteria	Record eligibility conditions
Application Deadline; Application Link	Support timely follow-up
Opportunity Status	Record open or closed status

The minimum CGPA is used in the inspected script. Automatic interpretation of the separate Eligibility Criteria text was not established.

4.4.1 Maintaining an Opportunity

The title and organisation distinguish the opportunity in the catalogue. Description, type and required skills explain its nature, while the deadline and application link help the user find the next step. Opportunity Status records a selected state such as open or closed.

The application owner should check the original opportunity notice before entering or updating a deadline. A minimum CGPA is meaningful only if its scale is understood and agrees with the student profile. The text eligibility field should also be reviewed, as it may contain conditions not covered by the script.

4.5 Related Forms

Table 4.4: Related forms

Form	Visible fields
Applications	Student, Opportunity, date, match percentage, missing skills, status and remarks
Opportunity Matches	Student, Opportunity, percentage, matching and missing skills, last calculated
Skill Development	Student, skill, related opportunity, priority, status, target date and notes
Student Certificates	Student, name, issuer, completion date, document text, extracted text and verification status

The Certificate Document and Extracted Text controls appeared as text areas; their labels do not establish upload or automatic extraction.

4.5.1 Interpretation of Related Records

An application records a student's relationship to an opportunity and can carry a status and remarks. A separate match record provides places for matching skills, missing skills and a percentage. Keeping those purposes separate allows the information to be reviewed without treating a match as an application.

A skill-development record can document a learning goal and target date, while a certificate record can describe an achievement and its verification status. The user must confirm the information; the visible field names alone do not establish an automated verification process.

4.6 Validation Flow

Step	Configured action
1	The Applications form is created or edited and submitted.
2	The action fetches the selected opportunity.
3	The script reads its minimum CGPA and the selected student's CGPA.
4	If the student value is lower, an alert is displayed and submission is cancelled.
5	Missing or incompatible values require execution testing.

The workflow was shown as enabled. Configuration indicates intended logic, not a successful end-to-end test.

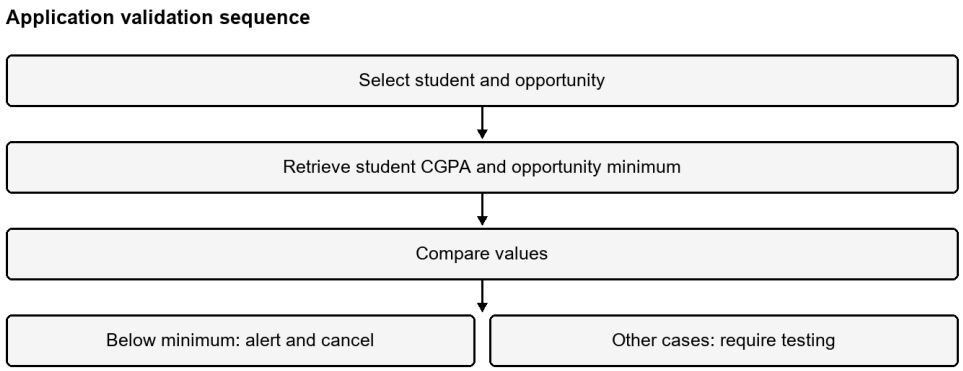


Figure 4.3: Configured CGPA validation sequence

4.6.1 Decision Boundary

The inspected condition uses a strict less-than comparison. For equal or higher values, this particular condition should not trigger the cancellation branch, but the actual submission outcome can be affected by other validation or missing data and must be tested.

If the two records use different grading scales, a numeric comparison can be misleading. The project should define a consistent scale before relying on this validation for a real application decision.

4.7 Navigation and Access

The six navigation groups each provide an entry form and an “All” report. The inspection used the owner account. Other roles and permissions were not tested.

Before wider use, review access to student contact and academic data. Establish a consistent CGPA scale, check opportunity deadlines and links, and review status information for accuracy.

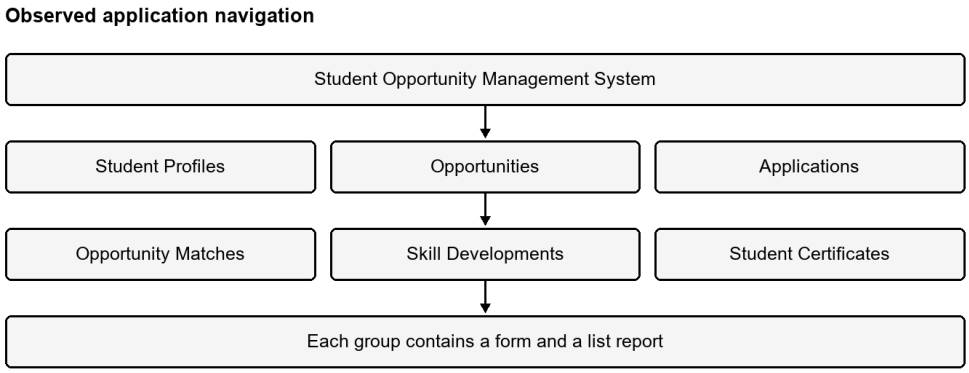


Figure 4.4: Observed navigation structure

4.7.1 Operational Responsibilities

An application owner should review who can enter, change and view records. Students may reasonably need access to their own information, while a coordinator may require a broader view; these are design considerations, not verified permission settings.

Reports should be checked periodically for outdated opportunities, inaccurate profile values and incomplete records. Any changes to field names or workflows should be followed by a regression test.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

Two similarly named applications were available in the account. The selected version has six form and report groups, including Opportunity Matches and Student Certificates, and therefore provides broader observed coverage.

The application uses text inputs, multiline fields, dates, dropdowns, radio buttons and lookup selections. Submit and Reset controls were visible on inspected forms. The Student Profiles report displayed six existing records during review; those records were not modified.

5.2 Evidence

The screenshots in this chapter are from the selected application and workflow builder. They show configuration and form appearance, not proof of complete submission or end-to-end operation.

5.2.1 Selection of the Project

Both available applications carried the same project name. The selected version was chosen because its navigation exposed six groups, compared with four in the other version. The additional groups are Opportunity Matches and Student Certificates.

The report follows the selected version’s visible interface. It does not merge data from the two applications or claim that a feature in one version necessarily exists in the other.

5.3 Component Inventory

Table 5.1: Component inventory

Form	Report
Student Profile	All Student Profiles
Opportunities	All Opportunities
Applications	All Applications
Opportunity Matches	All Opportunity Matches
Skill Development	All Skill Developments
Student Certificates	All Student Certificates

Reports display saved records and are accessible through the same navigation area as the corresponding form. Available actions for other users depend on their permissions.

5.3.1 Working with Reports

A form is the data-entry view; its related list report is the view used to inspect saved rows. A report may display multiple columns from the form. The owner should review filters and permissions when determining what another user will see.

The Student Profiles report was opened in the owner account and showed six rows at the time of inspection. That count is a point-in-time observation, not an adoption or performance measure.

5.4 Student and Opportunity Records

Student Profile groups contact details, department, year, CGPA, skills, summaries and career goals. Opportunities records the title, organisation, type, description, skills, minimum CGPA, eligibility, deadline, link and status.

These records provide selections for other forms. Consistent terminology and a common CGPA scale make the information easier to compare. Review stale information before using it in an application decision.

5.5 Reports and Navigation

List reports provide a structured view of saved data. The Student Profiles report showed existing rows and columns for profile fields. It was inspected without creating, modifying or deleting records.

The match and application forms include percentage and skill fields. The inspection did not establish whether their values are calculated automatically.

5.5.1 Record Quality

A profile and an opportunity can be linked through lookups rather than relying on a user to type the same identifying information repeatedly. This makes a relationship visible in the application, but the selected source record still needs to be correct.

The opportunity description and eligibility criteria complement the numeric minimum. A user should check the source notice, current deadline and any non-CGPA conditions when reviewing an application.

5.6 Deluge Validation

The enabled Validate CGPA Requirement workflow is associated with Applications, Created or Edited, and Validations on form submission. It contains one Deluge action. The script fetches the selected opportunity, reads its minimum CGPA and the selected student's CGPA, and cancels submission after an alert if the student value is lower.

A meaningful comparison depends on both records and values being present and expressed on the same scale. The test cases in Chapter 6 should be run before relying on this action.

5.7 Applications and Learning Records

The Applications form includes Application Date, Application Status and Remarks alongside student and opportunity selections. Skill Development records a skill to learn, related opportunity, priority, learning status, target completion date and notes.

These fields support human review. The exact permitted status values and transitions should be confirmed before developing a formal process around them.

5.7.1 Workflow Conditions

The observed action is attached to a validation event for a created or edited application. The script uses the selected opportunity to obtain the minimum and reads CGPA from the selected student. It then compares the values and cancels the submission if the student value is smaller.

The configured condition says the action executes without an additional workflow condition. The code itself still contains the CGPA comparison. The review did not submit records to observe the alert, cancellation or successful path.

5.8 Certificates and Matches

Student Certificates records a student selection, certificate name, issuer, completion date, document text, extracted text and verification status. The observed form presents the document and extracted text as text areas. The labels do not prove that a certificate has been uploaded, parsed or verified.

Opportunity Matches includes a percentage, matching and missing skills, and a last-calculated date. The source of those values and whether they update automatically were not verified.

5.8.1 Data Entry Considerations

The certificate form offers a place for an issuing organisation, completion date and verification status. These labels make it possible to record an assessment, but they do not show how a document is authenticated. Evidence should be checked by an authorised person.

Similarly, matching and missing skills can inform a learning plan only when their source is reliable. The report treats them as recorded fields rather than claiming that a recommendation engine generated them.

5.9 Implementation Evidence

The following pages show the actual form interfaces and the CGPA workflow. Screenshots capture the visible portion of each page; longer forms continue below the visible area.

5.10 Review Method

The signed-in owner interface was used to inspect navigation, form labels, an existing student report and the workflow configuration. No test records were submitted and no application settings were changed.

5.10.1 Reading the Screenshots

The screenshots are genuine views from the selected project. Each caption identifies the component being shown. Form screenshots capture a visible portion of the page; where a form extends beyond the browser viewport, the remaining fields are described in the corresponding design tables.

The workflow screenshot is evidence of configuration, not a test result. No test data was submitted or changed during the inspection used for these images.

5.11 Student Profile

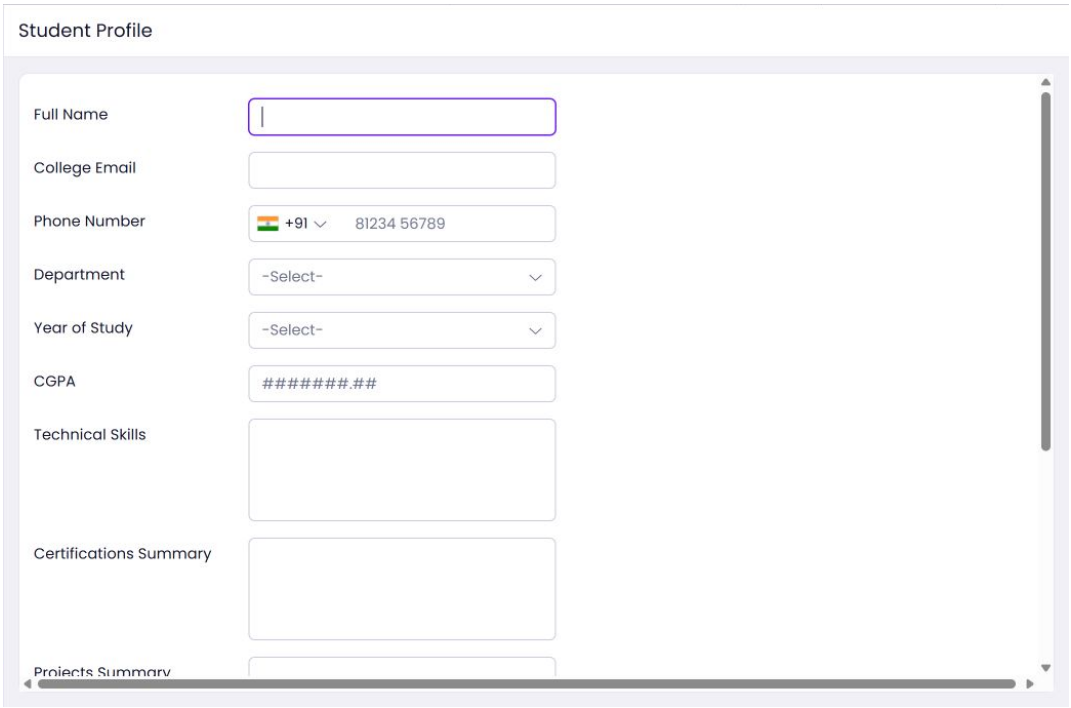


Figure 5.1: Student Profile form

The upper portion collects identity, department, year, CGPA and technical skills. Additional fields continue below.

5.11.1 Field Interpretation

The profile form brings together identity, educational context and skill information. The screenshot shows the upper portion of the form; project summaries, career goal and profile status appear further down. A user should review the selected grading scale before entering CGPA.

5.12 Opportunities

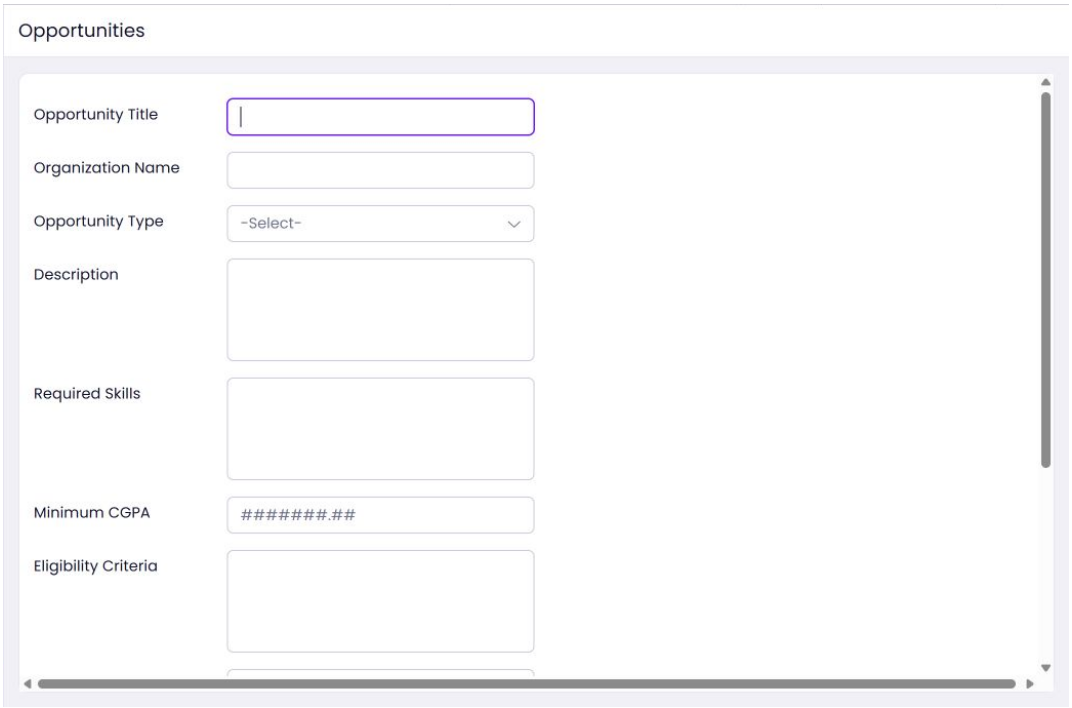


Figure 5.2: Opportunities form

The upper portion records the title, organisation, description, required skills and minimum CGPA.

5.12.1 Field Interpretation

The opportunity form records a title and organisation together with a description, required skills and minimum CGPA. Other fields further down include eligibility criteria, deadline, application link and status. The source notice should be checked whenever these details are updated.

5.13 Applications

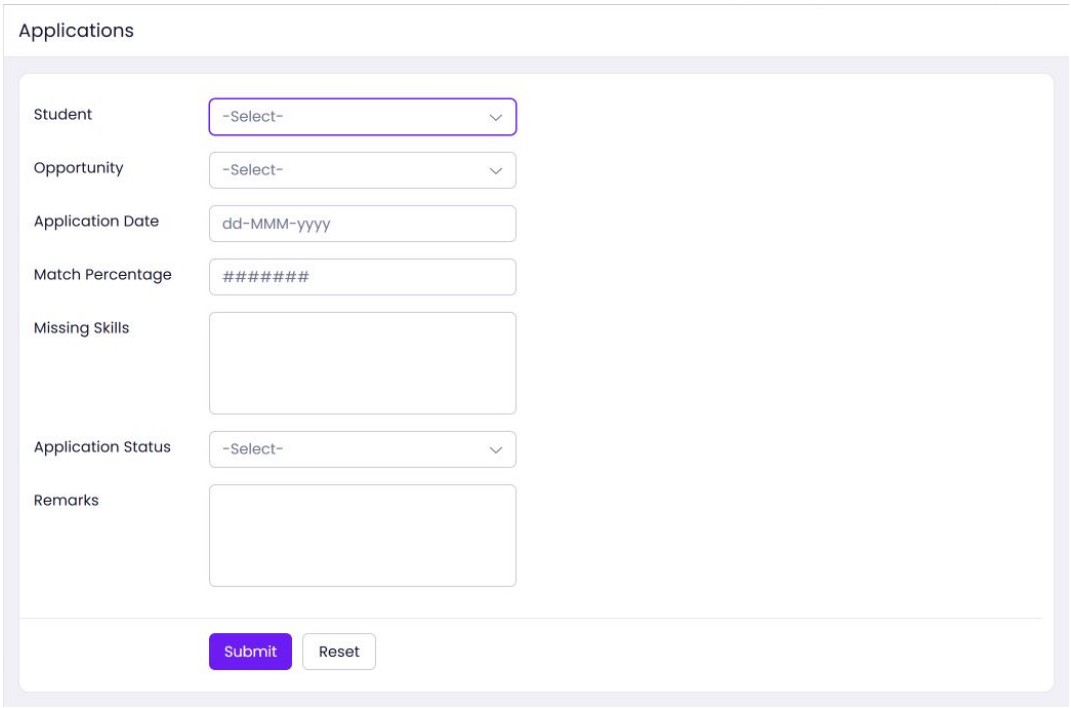


Figure 5.3: Applications form

The form associates a student and opportunity with date, match details, status and remarks.

5.13.1 Field Interpretation

The application form links a student and opportunity using dropdown selections. It also provides a date, match percentage, missing skills, status and remarks. The visible fields describe data entry; the screenshot does not demonstrate successful submission.

5.14 Skill Development

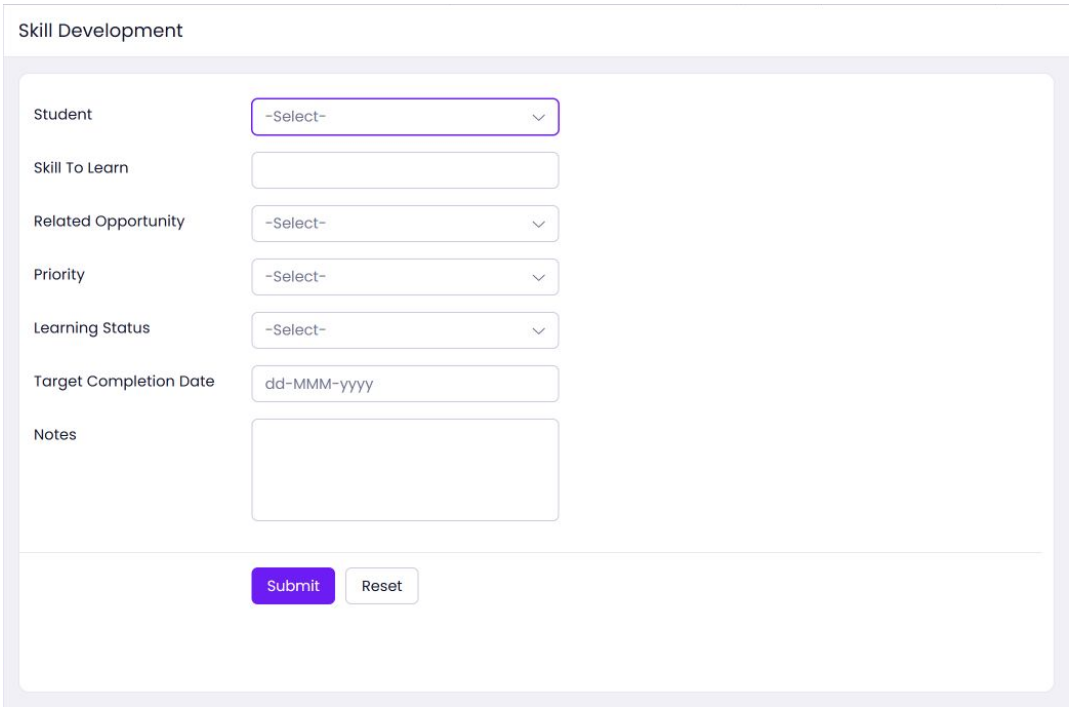


Figure 5.4: Skill Development form

The form records a learning goal, opportunity, priority, status, target date and notes.

5.14.1 Field Interpretation

The Skill Development form captures the skill to learn and optionally relates it to an opportunity. Priority, learning status, target date and notes can support a human follow-up. Their values should be reviewed and updated as the student progresses.

5.15 Student Certificates

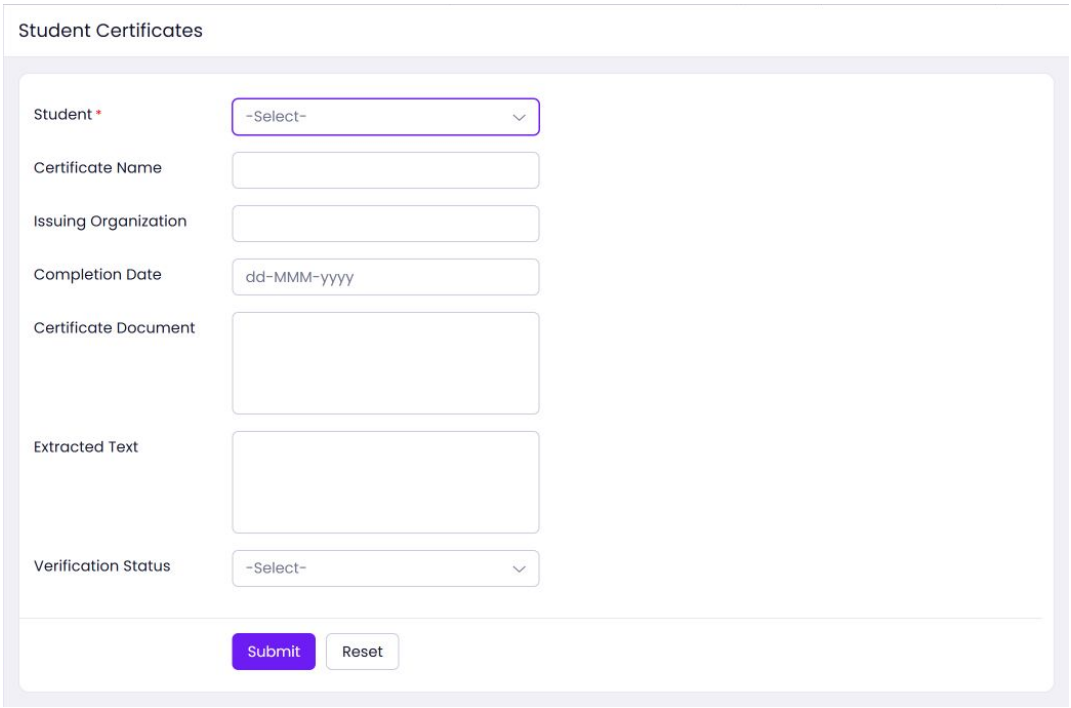


Figure 5.5: Student Certificates form

The form associates certificate details and verification status with a student.

5.15.1 Field Interpretation

The certificate form records a selected student, certificate name, issuer and completion date. Its document and extracted-text controls appear as text areas, and a verification-status selection is available. Automatic extraction and verification were not established.

5.16 Opportunity Matches

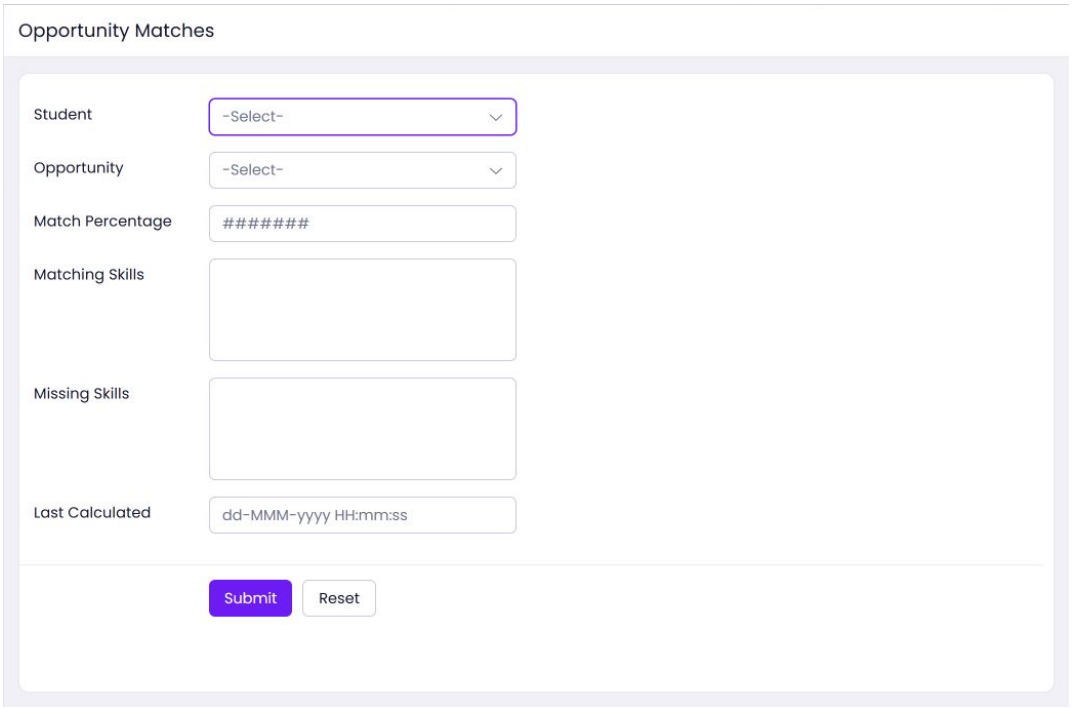


Figure 5.6: Opportunity Matches form

The form provides fields for student, opportunity, percentage, skills and calculation date.

5.16.1 Field Interpretation

A match record associates a student and an opportunity with percentage and skill fields. The last-calculated field is visible, but the review did not establish a calculation method or determine whether the date is updated automatically.

5.17 CGPA Validation Workflow

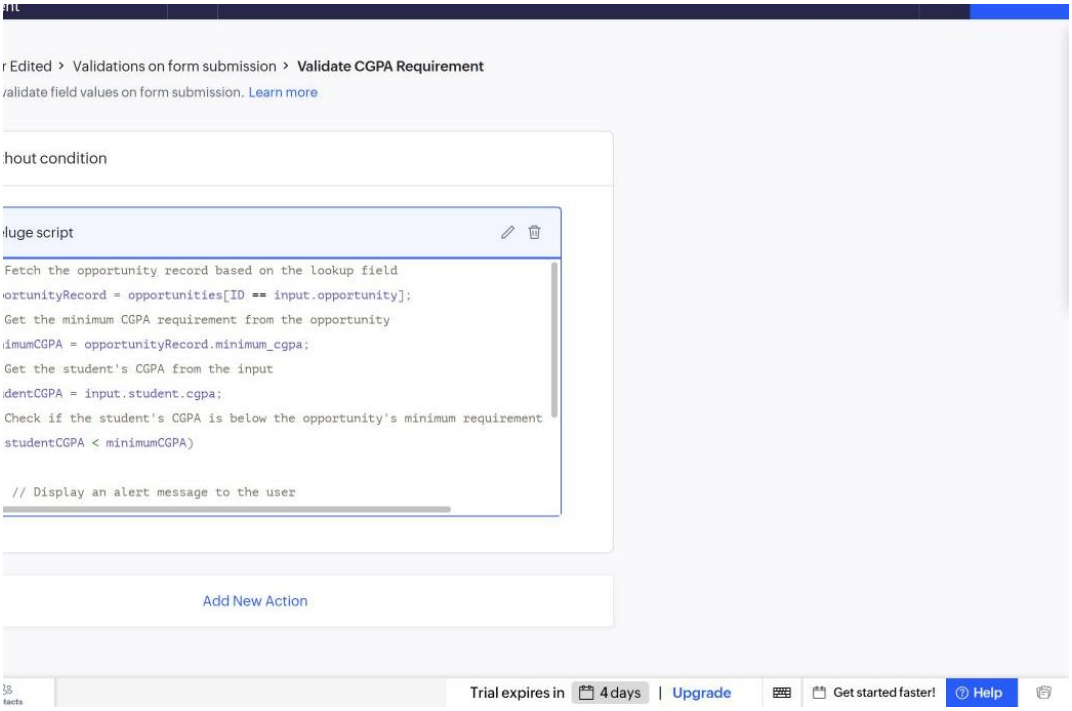


Figure 5.7: CGPA validation workflow

The builder shows the Deluge action for application validation. Controlled runtime tests remain necessary.

5.17.1 Configuration Interpretation

The displayed action is part of the Applications validation workflow. Its code reads the opportunity minimum and student CGPA. A test should verify the alert and cancellation as well as the equal, higher and missing-value cases.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Test Strategy

The tests below are proposed for a controlled environment. Form navigation and the configuration were observed; submission scenarios were not executed. Expected outcomes are not recorded as passes.

Table 6.1: Proposed test cases

ID	Scenario	Expected result	Status
T01	Open six forms	Correct form appears	Observed
T02	Open student report	Saved rows appear	Observed
T03	CGPA below minimum	Alert; submission cancelled	Pending
T04	CGPA equal to minimum	Less-than check does not reject	Pending
T05	CGPA above minimum	Less-than check does not reject	Pending
T06	Missing student or CGPA	Safe and clear handling	Pending
T07	Missing opportunity minimum	Safe and clear handling	Pending

6.1.1 Test Evidence

A test record should include the scenario, input values, expected result, actual result, date and evidence. An observed interface or script is useful supporting evidence, but it cannot substitute for an executed test case.

The below-minimum scenario should verify both the message and the absence of an added application record. For equal and above values, the reviewer must also account for any other form validations that may prevent saving.

6.2 Test Procedure

In an authorised test environment, create a student and opportunity using the same CGPA scale. Record the values, select both in an application and observe the result. Repeat below, equal and above the minimum. Record the actual output and any error message.

6.3 Additional Cases

- Check missing selections and blank values.
- Check grading scales and define one consistent policy.
- Review date, status and remarks after a successful submission.
- Test permissions separately for each intended user role.
- Verify any claimed calculation or extraction with direct evidence.

6.3.1 Boundary Values

A useful boundary set uses one value just below the opportunity minimum, one equal value and one value just above it. Use the same scale for both records and document the selected opportunity and student. Then repeat with missing values to determine whether the user receives a clear response.

Test data should be labelled and handled according to the owner's normal cleanup process. A test should not overwrite an existing student's real academic information merely to produce a convenient result.

6.4 Review Log

Observation	Next step
CGPA validation configured	Run controlled boundary tests
Missing-value behaviour unknown	Test incomplete records
CGPA scales may differ	Agree on and validate a common scale
Match and extracted text fields present	Verify how values are produced
Separate role permissions not reviewed	Test representative roles

These are review items, not confirmed defects. A defect should be supported by reproducible steps, expected and actual results, and evidence.

6.4.1 Recording a Defect

When a test fails, capture the form, relevant inputs and the exact result without exposing unnecessary personal information. Record whether the issue is reproducible and whether another validation or access rule could explain the outcome.

After a correction, repeat the original failing scenario and the adjacent boundary cases. Keep configuration observations separate from confirmed runtime defects.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Deployment Considerations

The application was accessed through an owner account. Wider deployment requires a review of the audience, account subscription, data ownership and access settings. A separate tested production environment was not established.

Table 7.1: Deployment checklist

Check	Action
Access	Review users and permission sets
Data	Confirm test data and CGPA scale
Logic	Run representative validation tests
Operations	Review subscription and availability
Recovery	Establish backup and recovery arrangements
Release	Record approval and test evidence

7.1.1 Release Preparation

The owner should identify which records are demonstration data and which are intended for continued use. A release review should confirm the intended users and the actions they can perform, especially where the application contains student contact and academic information.

Account status and the Creator environment are dependencies. The report does not claim a tested production release, service-level agreement or completed recovery exercise.

7.2 Security and Privacy

Profiles contain contact and academic information. Limit access to people who need it, review sharing settings, and keep passwords and authentication codes out of records and project documentation. Certificate text may contain sensitive information and should be reviewed before entry.

7.3 Data Quality and Maintenance

Use a consistent grading scale, maintain deadlines and links, and periodically review statuses. Retest dependent forms after a workflow or field change and keep a recovery plan suitable for the data.

7.3.1 Ongoing Review

An opportunity may expire or change after it is entered. A named owner should periodically review deadlines, statuses and source links. Profiles should be corrected by an authorised user when academic or skill information changes.

When application settings or scripts change, retain a description of the change and rerun relevant tests. Backup and restore procedures should be documented and tested according to the needs of the account and the sensitivity of its data.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Observed Output

The selected application presented six groups with form and report links. The inspected forms exposed the fields described in Chapter 4. The Student Profiles report contained six rows at inspection time. The workflow builder displayed an enabled application validation action with a CGPA comparison.

Screenshots illustrate the interface and configuration. No records were created during this review, and no success or failure of the scripted comparison was observed at runtime.

8.1.1 What the Evidence Shows

The forms demonstrate the structure available for data entry. The report view demonstrates that existing student rows can be displayed in the owner session. The workflow builder demonstrates that a validation action is configured. These observations support an explanation of design and configuration, rather than a claim of complete operational success.

The selected application has more visible areas than the other same-named project. That comparison guided selection for this report, but does not measure quality, accuracy or use by students.

8.2 Objectives and Evidence

Table 8.1: Objectives and evidence

Objective	Evidence	Remaining work
Student data	Profile form and report visible	Review data quality
Opportunities	Form and report link visible	Check dates and sources
Applications	Form and report link visible	Run submission tests
Learning and certificates	Forms visible	Verify processes
CGPA check	Enabled action inspected	Test boundary and missing values
Match details	Form visible	Verify calculation method

8.2.1 Assessment Method

Each objective is assessed against direct observations made in the application. Where a component was visible, the corresponding entry is described as observed; where execution was not attempted, the remaining work is stated explicitly.

This approach makes it possible to plan the next demonstration without assuming that a field label is a working automation or that a configured workflow has passed its boundary tests.

8.3 Discussion and Limitations

The project brings related activities into one interface. Separate forms give each record a clear purpose, and lookup selections associate information with a student or opportunity. Reports support review of saved data.

The read-only review does not demonstrate successful application submission, complete role testing, automatic matching, certificate extraction, notifications or production readiness. Existing records do not prove every workflow has succeeded.

8.4 Future Work

Priorities include controlled validation tests, consistent grading scales, access review, useful report filters and verification of any matching or certificate-processing features.

8.4.1 Practical Next Steps

A useful next demonstration would use a small, clearly identified set of test records, consistent CGPA values and a documented expected result. The reviewer can then verify the below-minimum alert and compare the final report with the inputs.

Separate tests should examine who can see each report and how stale opportunities are managed. Any automated matching or certificate extraction must be documented from its actual configuration and validated independently.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Student Opportunity Management System demonstrates a low-code design for maintaining student, opportunity, application, skill, certificate and match information. The inspected CGPA validation illustrates how a Creator form event can invoke Deluge logic using related records.

9.2 Future Enhancements

The next iteration should prioritise controlled tests, consistent grading scales, missing-value handling, role-specific access, audit records and recovery planning. Any automated matching or extraction should have a defined method and verified results.

9.3 Learning Outcomes

The project combines requirement analysis, form design, related data, reports, event-driven scripting, privacy and test planning in an academic scenario.

9.4 Final Observation

The project's strongest demonstrated elements are the linked form structure, the owner's report view and the configured validation logic. Completing the proposed tests would establish whether the intended workflow behaves as expected and identify any gaps in data or permissions.

REFERENCES

1. Zoho Creator, product documentation, https://www.zoho.com/creator/help/.

2. Zoho Creator, Forms and Reports, https://www.zoho.com/creator/.

3. Zoho Creator, workflow documentation, https://www.zoho.com/creator/help/.

4. Zoho Deluge, scripting documentation, https://www.zoho.com/deluge/help/.

Application-specific details are based on the inspected Creator interface and configuration. General documentation does not certify this application's runtime behaviour.

Use of References

The cited platform documentation provides context for Creator forms, reports, workflows and Deluge. The observations about this particular application come from its inspected user interface and builder configuration; the references are not a substitute for project-specific testing.

APPENDIX A — ZOHO CREATOR FEATURE COVERAGE

Table A.1: Feature coverage

Feature	Evidence	Status
Forms	Six form areas	Observed
Reports	Six report links	Observed
Lookup controls	Student and opportunity selections	Observed
Date and status fields	Multiple forms	Observed
Workflow	Applications CGPA validation	Configuration observed
Deluge	Comparison and cancel submission	Source observed
Record entry	Submit buttons visible	Not tested
Permissions	Owner account used	Other roles untested

Checklist Interpretation

Observed means the component or setting was visible during the review. It does not mean all paths through it were tested. A configured action should be exercised using the scenarios in Chapter 6 before it is relied on operationally.

Feature	Evidence	Status
Automatic matching	Match fields present	Not established
Certificate extraction	Extracted Text field present	Not established
Email alerts	No relevant configuration inspected	Not established
Mobile use	Device-view controls visible	Not tested
Production release	No separate release evidence	Not established
Backup and restore	No restore test inspected	Not established
Report filtering	Report views present	Not fully tested

A field or platform capability should not be confused with a verified end-to-end feature. This checklist provides a starting point for further review.

Further Review

Unestablished items may be implemented or configured elsewhere in the project, but they were not established by this inspection. A subsequent review should inspect the relevant configuration and record a reproducible outcome before changing their status.

APPENDIX B — COURSE OUTCOME MAPPING

Table B.1: Course outcome mapping

Learning area	Application
Requirements	Identify stakeholders, scope and needs
Design	Organise related forms and reports
Implementation	Configure fields and a workflow
Scripting	Explain the CGPA comparison
Testing	Design positive, boundary and missing-value cases
Governance	Consider access and data quality
Communication	Present evidence and limitations

Learning Evidence

The project demonstrates the relationship between a requirement, a form field, a report and a validation action. The proposed tests provide an opportunity to compare the intended result with actual behaviour and to document a correction if they differ.

APPENDIX C — DELUGE SOURCE AND CONFIGURATION

C.1 Workflow Configuration

Table C.1: Configuration

Property	Observed setting
Form	Applications
Workflow	Validate CGPA Requirement
Event	Created or Edited
Form event	Validations on form submission
Status	Enabled
Actions	One Deluge script
Condition	Execute without condition

The following source is transcribed from the visible action. It is presented as configuration without a claim that every execution path has been tested.

C.1.1 Event Context

The event is validation on application submission when a record is created or edited. The builder shows one action under an execute-without-condition block; the action's own script contains the conditional comparison. Enabled status describes configuration at inspection time.

C.2 Validation Script

```
opportunityRecord = opportunities[ID == input.opportunity];
minimumCGPA = opportunityRecord.minimum_cgpa;
studentCGPA = input.student.cgpa;

if (studentCGPA < minimumCGPA)
{
    alert(opportunity, "Student CGPA is below the minimum requirement for this
opportunity.");
    cancel submit;
}
```

The action reads the minimum and student values. A less-than comparison triggers an alert and cancels submission. Comments are omitted here for readability; the workflow screenshot shows the editor.

C.2.1 Script Review

The opportunity lookup is used to fetch the related record. Its minimum CGPA is compared with the CGPA reached through the student lookup. The alert and cancel statement appear inside the less-than branch. The script should be tested against missing values and the correct grading scale.

C.3 Source Interpretation

Item	Meaning
input.opportunity	Selected opportunity
opportunities[ID == input.opportunity]	Fetch the corresponding record
minimum_cgpa	Opportunity minimum
input.student.cgpa	CGPA on the selected student record
alert(...)	Message explaining the rejection
cancel submit	Stop the current form submission

Missing values and incompatible grading scales require explicit tests. Source configuration alone does not prove these cases are handled correctly.

C.3.1 Inputs and Outcomes

The action depends on valid selected records and numeric values that can be compared meaningfully. The expected rejection path is documented in the source, but the review did not observe the alert or the absence of a saved record during execution.

If the values are equal, the shown less-than condition is false. This observation concerns the displayed condition only; other validation may still affect the submission.

C.4 Configuration Reference

Component	Key controls
Student Profile	Name, email, department, year, CGPA and skills
Opportunities	Title, organisation, skills, minimum CGPA and deadline
Applications	Student and opportunity, date, status and remarks
Opportunity Matches	Lookups, percentage, skills and calculation date
Skill Development	Learning goal, priority, status and target date
Student Certificates	Student, issuer, completion, text and verification status

C.4.1 Configuration Maintenance

When a label, field link name or lookup relationship changes, review any script that refers to it. Maintain a short change note and repeat the relevant submission tests. The field inventory describes controls seen in the owner session and is not a complete schema export.

C.5 Verification Notes

Applications, Student Certificates and Student Profile appeared in the builder's Forms with workflows list. Only the Applications validation was opened and inspected in detail. The contents and runtime outcomes of other actions are outside this report's verified evidence.

C.6 Recommended Test Evidence

- Record the test date, inputs and expected outcome.
- Capture an alert or successful record in an authorised environment.
- Record the actual result and any error message.
- Repeat after changing a relevant field, script or permission.

C.6.1 Evidence Handling

Use fictitious or approved sample values in test evidence. Preserve the expected and actual outcomes together so that a reviewer can understand the result without searching unrelated records. Avoid displaying private student contact information in screenshots unless it is necessary and authorised.

APPENDIX D — USER MANUAL

D.1 Using the Application

- Sign in to the authorised Creator account and open Student Opportunity Management System.
- Use Student Profiles to enter or review academic and skill details.
- Use Opportunities to record requirements and deadlines.
- Open Applications, select a student and opportunity, and enter date, status and remarks.
- Review any eligibility message and correct source data where appropriate.
- Use Skill Developments, Student Certificates and Opportunity Matches for their records.
- Open each All report to review saved entries; check access before sharing data.

Test the configured validation with authorised sample records before relying on it for decisions.

D.2 Before Submitting

Confirm that the selected student and opportunity are the intended records. Review the student's CGPA, the opportunity's minimum and the relevant eligibility text. If the app shows an alert, check the source data and do not alter values merely to bypass a legitimate requirement.

After a successful submission, open the corresponding report and confirm that the record appears with the intended status. This is a proposed user procedure and should be verified in an authorised test environment.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Vidyut Rao S	Register No.	240701587
Project Title	Student Opportunity Management System	Date	_____

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.

Level	% of Max. Marks	Descriptor
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Vidyut Rao S	Register No.	240701587
Department / Section	Computer Science and Engineering Section: CSE-J	Date of Viva	_____
Application URL Verified	Yes / No	Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____